10.2 DC: Potential Dividers

Question Paper

Course	CIE A Level Physics
Section	10. D.C. Circuits
Topic	10.2 DC: Potential Dividers
Difficulty	Easy

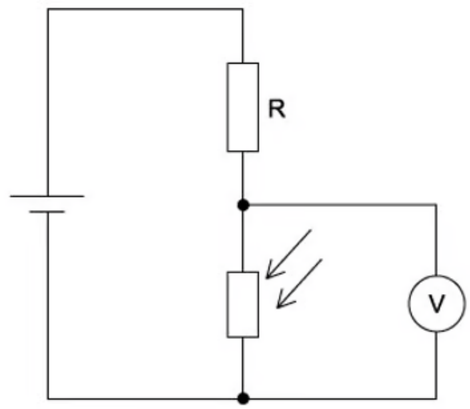
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A potential divider consists of a fixed resistor R and a light-dependent resistor (LDR)



What happens to the voltmeter reading, and why does it happen, when the intensity of light on the LDR increases?

A. The voltmeter reading decreases because the LDR resistance decreases

B. The voltmeter reading decreases because the LDR resistance increases

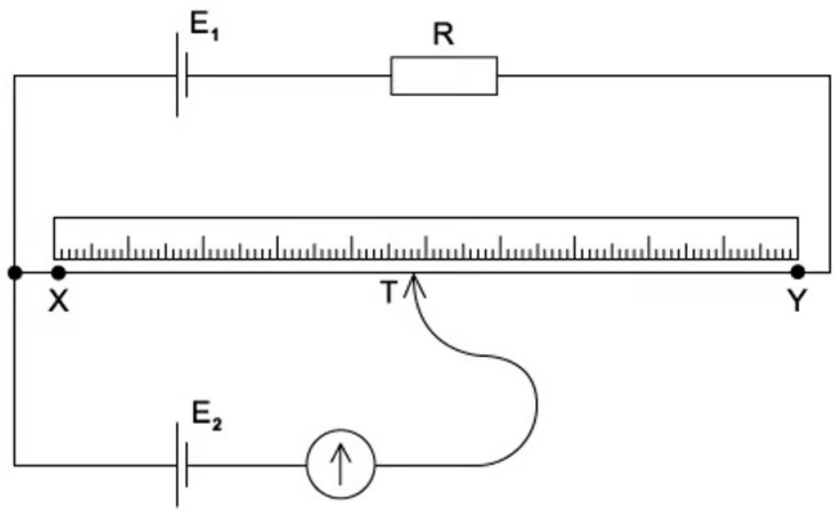
C. The voltmeter reading increases because the LDR resistance decreases

D. the voltmeter reading increases because the LDR resistance increases

[1 mark]

Question 2

The diagram shows a potentiometer circuit.



The contact T is placed on the wire and moved along the wire until the galvanometer reading is zero. The length XT is then noted.

In order to calculate the potential difference per unit length of the wire XY, which value must also be known?

A. the e .m .f. of the cell E_1

B. the e. m .f. of the cell E_2

C. the resistance of resistor R

D. the resistance of the wire XY

[1 mark]

Question 3

A source of e.m.f. 9.0 mV has an internal resistance of 6.0 Ω.

It is connected across a galvanometer of resistance 30 Ω.

What is the current in the galvanometer?

A. 250 μA

B. 300 μA

C. 1.5 mA

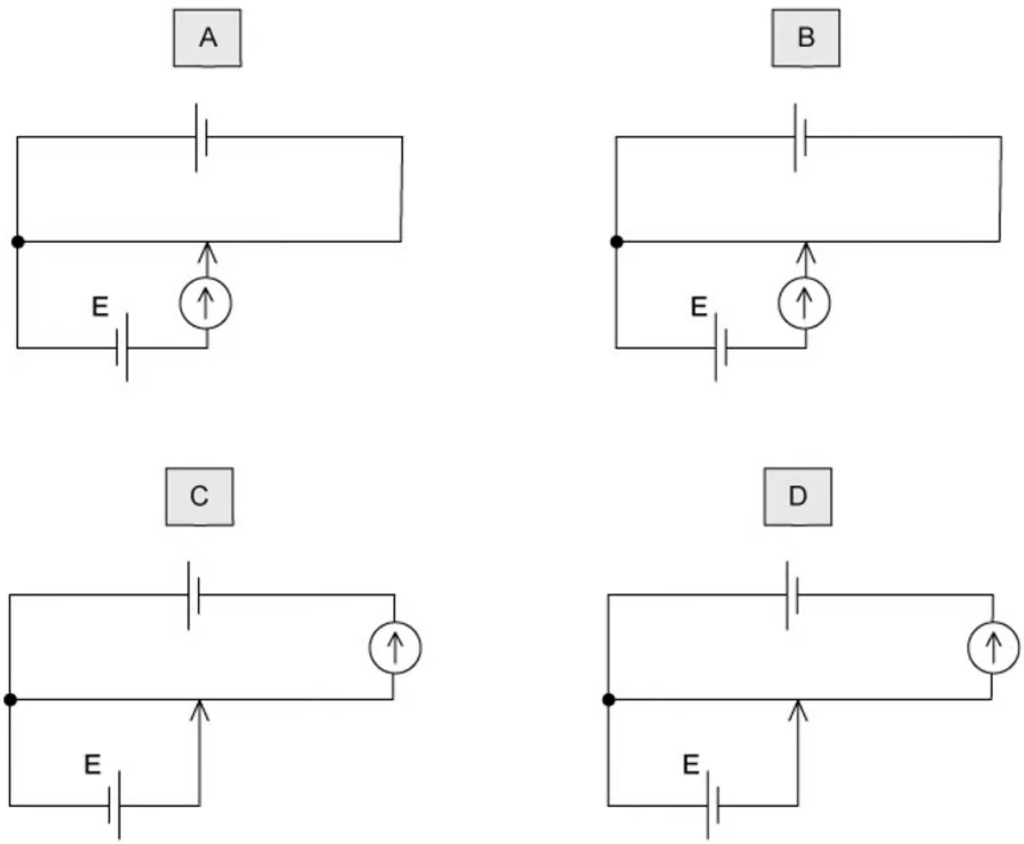
D. 2.5 mA

[1 mark]

Question 4

The unknown e.m.f. *E* of a cell is to be determined using a potentiometer circuit. The balance length is to be measured when the galvanometer records a null reading.

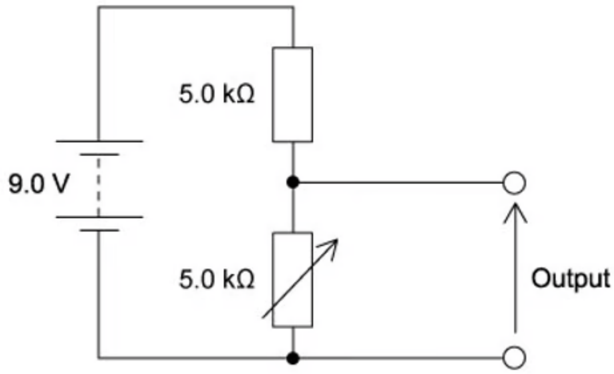
What is the correct circuit to use?



[1 mark]

Question 5

The diagram shows a potential divider circuit designed to provide a variable output p. d.



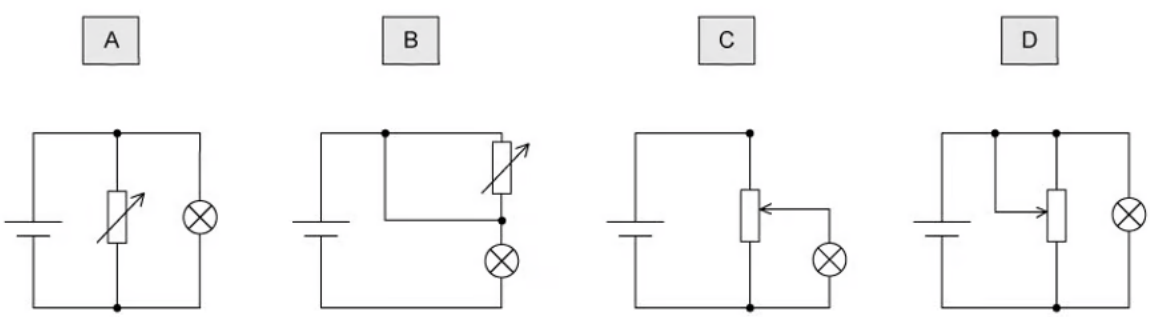
Which gives the available range of output p. d?

	Maximum output	Minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V

[1 mark]

Question 6

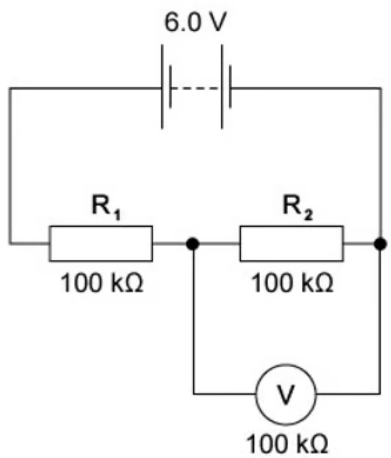
Which diagram shows a potential divider circuit that can vary the voltage across the lamp?



[1 mark]

Question 7

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter have resistance 100 kΩ.



What is the current in the resistor R_2?

A. 20 μA

B. 30 μA

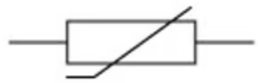
C. 40 μA

D. 60 μA

[1 mark]

Question 8

Which electrical component is represented by the following symbol ?



A. a diode

B. a potentiometer

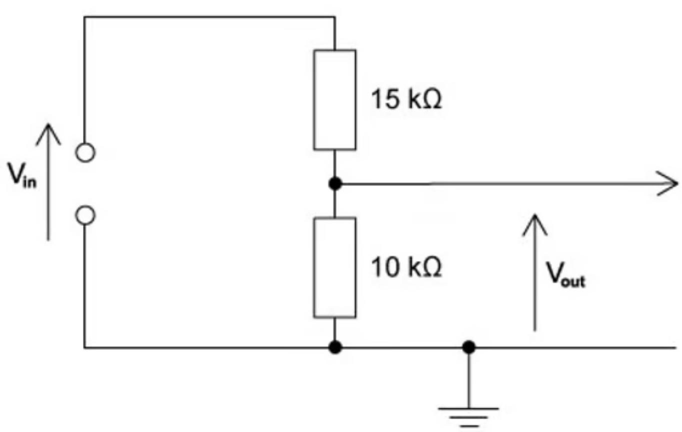
C. a galvanometer

D. a thermistor

[1 mark]

Question 9

The circuit is designed to trigger an alarm system when the input voltage exceeds some preset value. It does this by comparing V_{out} with a fixed reference voltage, which is set at 4.8 V .



V_{out} is equal to 4.8 V.

What is the input voltage V_{in}?

A. 4.8 V

B. 7.2 V

C. 9.6 V

D. 12 V

[1 mark]

Question 10

A low-voltage supply with an e. m. f. of 20 V and an internal resistance of 1.5 Ω is used to supply power to a heater of resistance 6.5 Ω in a fish tank.

What is the power supplied to the water in the fish tank ?

A. 41 W

B. 50 W

C. 53 W

D. 62 W

[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- The resistance of the LDR decreases as the light intensity increases
- Ohm's law states $V = IR$
- Since the resistance of the LDR decreases, the potential difference across it also decreases
- Therefore, as the LDR resistance decreases, the voltmeter reading decreases

2

The correct answer is **B** because:

- When the galvanometer reading is zero, this means that the p.d. across XT of the wire is the same as the e.m.f. E_2 connected to the galvanometer
- To calculate the potential difference per unit length on the wire XY, we need to know the length of the wire, and the p.d. across it
- The p.d. across XT is equal to the e.m.f. E_2 when the galvanometer reading is zero
- So, we need to know the e. m .f. of the cell E_2 to know the p. d. across the wire

3

The correct answer is **A** because:

- The galvanometer and the internal resistance of the cell can be treated as two resistors in series
- So, the total resistance R in the circuit = 6 + 30 = 36 Ω
- Ohm's law states $V = IR$
    - So, current $I = \frac{V}{R} = \frac{9.0 \times 10^{-3}}{36} = 2.5 \times 10^{-4}\ A = 250 \times 10^{-6}\ A$
- In a series circuit, the same current flows through the cell and the galvanometer
    - Therefore, the current in the galvanometer is 250 μA

4

The correct answer is **B** because:

- For the galvanometer to give a null reading, the potential difference across it should be zero
- This can only occur when the position at which it is connected on the wire gives a potential difference (on the wire) equal to the cell of e .m .f. *E*
- The cell should be connected such that its potential opposes the potential on the wire
- When the potential difference across the galvanometer is zero, no current flows through it, and the galvanometer records a null reading
    - This set up is demonstrated by circuit **B**

A is incorrect because current flows from the positive terminal to the negative terminal, so, the positive terminal of the cell cannot be connected directly to the galvanometer because this will always cause a current to flow through it

C & **D** are incorrect because the galvanometer cannot be connected directly to the negative terminal of the main supply because all the current would return to the negative terminal of the main supply, causing the galvanometer always to give a non-zero reading

5

The correct answer is **B** because:

- The potential divider equation is given by
    - $V_{out} = V_{in} \times \left(\frac{R_1}{R_1 + R_2} \right)$
    - Where R_1 = 5 kΩ, R_2 = 5 kΩ and V_{in} = 9.0 V
    - $V_{out} = 9 \times \left(\frac{5}{5+5} \right) = 4.5\ V$
- Therefore, the minimum output of the variable resistor is 0 V, and the maximum output is 4.5 V

6

The correct answer is **C** because:

- A variable potential divider (potentiometer) can be used to vary the brightness of a light bulb between zero and normal brightness
- The moveable contact, represented by the arrowhead, splits the resistor in two allowing it to act as though there were two resistors in the circuit and hence like a variable potential divider or potentiometer
- Changing the resistance of the potentiometer changes the potential difference of the lamp
- This is represented by circuit **C** as the lamp needs to be directly connected to the potentiometer

A & **B** are incorrect because a potentiometer is represented by an arrow pointing to a resistor

D is incorrect because the wire with the arrowhead needs to connect the lamp directly to the potentiometer in order to vary the potential difference of the lamp

7

The correct answer is **A** because:

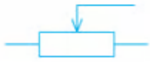
- We can treat the three components as two sets of components in series (R_1 and the parallel combination of R_2 and the voltmeter)
- So, the total resistance in the circuit $= 100 + \left[\frac{1}{100} + \frac{1}{100}\right]^{-1} = 100 + 50 = 150\ k\Omega$
- Ohm's law states $V = IR$
  - Therefore, the total current in circuit $I = \frac{6.0}{150} = 40\ \mu A$
- This current of 40 μA flows through resistor R_1, but it splits at the junction of R_2 and the voltmeter, which are in parallel
- Since both have the same resistance, the current splits equally
    - So, the current in $R_2 = \frac{40}{2} = 20\ \mu A$

8

The correct answer is **D** because this symbol represents a thermistor

A is incorrect because the symbol for a diode is

B is incorrect because the symbol for a potentiometer is



C is incorrect because the symbol for a galvanometer is

9

The correct answer is **D** because:

- The potential divider equation is given by
- $V_{out} = V_{in} \times \left(\frac{R_1}{R_1 + R_2} \right)$
  - Where $R_1 = 10$ kΩ, $R_2 = 15$ kΩ and $V_{out} = 4.8$ V
- $V_{in} = V_{out} \div \left(\frac{R_1}{R_2 + R_2} \right) = V_{out} \times \left(\frac{R_1 + R_2}{R_1} \right)$
- $V_{in} = V_{out} \left(\frac{R_1 + R_2}{R_1} \right) = \frac{4.8(10 + 15)}{10} = 12\ V$
- Therefore, the input voltage is 12 V

10

The correct answer is **A** because:

- The power supplied to the heater is $P = VI = \frac{V^2}{R}$
- From the potential divider equation:

C is incorrect because the symbol for a galvanometer is

9

The correct answer is **D** because:

- The potential divider equation is given by
- $V_{out} = V_{in} \times \left(\frac{R_1}{R_1 + R_2} \right)$
  - Where $R_1 = 10$ kΩ, $R_2 = 15$ kΩ and $V_{out} = 4.8$ V
- $V_{in} = V_{out} \div \left(\frac{R_1}{R_2 + R_2} \right) = V_{out} \times \left(\frac{R_1 + R_2}{R_1} \right)$
- $V_{in} = V_{out} \left(\frac{R_1 + R_2}{R_1} \right) = \frac{4.8(10 + 15)}{10} = 12\ V$
- Therefore, the input voltage is 12 V

10

The correct answer is **A** because:

- The power supplied to the heater is $P = VI = \frac{V^2}{R}$
- From the potential divider equation:
  - $V_{heater} = V_{e.m.f} \times \left(\frac{R_1}{R_1 + R_2} \right)$
  - $V_{heater} = 20 \times \left(\frac{6.5}{6.5 + 1.5} \right) = 16.25\ V$
- So, $P = \frac{V^2}{R} = \frac{(16.25)^2}{6.5} = 40.625\ W$
- Therefore, the power supplied to the heater is 41 W